

Georgia Precinct Returns and the 2020 Recount

Georgia's precinct returns, the vote-method columns, and what happens when a state counts the same ballots twice

84-355 Democracy's Data

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The most precise public record of an American election is a set of files exported by county clerks. Georgia's has 2,656 rows in it, one per precinct, and the precincts are identified the way the counties identify them: "1B", "2", "Senior Center". There is no code, no stable key, nothing that would let a computer match a precinct in this file to the same ground four years later.

That is not a formatting complaint. It is the shape of the thing. A **precinct** is the building block of American elections — a few hundred to a few thousand voters, and the level at which ballots are physically counted before anything is added up. Everything coarser, from county returns to state totals to congressional district numbers, is a sum of precincts. And yet this file was not produced by an authority. It was produced by 159 separate county election offices, each exporting its own tabulation, and the state's contribution was to publish all 159 of them.

Two questions follow. What can be seen at this resolution that cannot be seen at any coarser one? And how much can a number this small be trusted?

01 Why the finest grain is the one that matters

Precincts are the only election geography fine enough to sit next to census blocks, which makes them the unit on which almost every consequential statistical claim about race and voting is built. Racially polarized voting analysis in Voting Rights Act litigation, redistricting plan evaluation, minority-opportunity district design — all of it runs on precinct returns joined to block demographics. When a court decides whether a map dilutes a minority group's vote, it is reading numbers that started here.

They are also where the **mechanics** of voting become visible. Whether a ballot arrived by mail, at an early-voting site, or in person on election day is recorded per precinct in Georgia. In 2020 that split was the single most contested fact in American politics.

02 How this data reaches the public is itself the argument

No level of American government publishes a machine-readable national precinct file. What exists instead is a sequence of substitutes, and this chapter has been through three of them.

It ran first on **MEDSL**, the MIT Election Data and Science Lab's compilation. Then on **VEST**, the Voting and Election Science Team. That is a volunteer academic project that reads state and county returns by hand, reconciles them, matches precincts to boundaries and publishes clean national files. VEST's work is careful, and even its column names are worth teaching.

But it is served through Harvard's Dataverse behind a **guestbook**: a form naming who you are and what you want the file for. A person with a browser satisfies it in thirty seconds. A script gets a sentence back where the file should be: "You may not download this file without the required Guestbook response for guestbookID 458."

It now reads neither. Everything here comes from the **Georgia Secretary of State's own publication**. That is the XML archives the state posts for the 2020 general, one ZIP per county, all 159 of them. And it is the JSON export of the 2024 general, from the state's newer results platform. Those are the counties' tabulation exports – the primary source, not a reconstruction of one. The course now has no guestbooked sources at all, down from three.

Be exact about what gating is left, because it is easy to overstate. On `sos.ga.gov` the address is stable and public, and a browser fetches the archive without complaint. But **a script gets refused on the identical address, with identical headers and a referer**. That was tested rather than assumed, and recorded in `data/README.md`.

So the 2020 archive is one manual download, once. The 2024 export is *not* gated: it sits on a CDN path that a script fetches unattended. The same state, four years apart, protected one way and not the other, with no announcement either time. What did not change is the shape of the publication. **Georgia posts 159 county files, not one state file**. Amos, McDonald and Watkins (2017) describe what assembling the national version of that costs, and the short answer is years.

Going upstream was not only a matter of independence. It bought a column. VEST publishes precinct by candidate. The state's own exports carry precinct by candidate by **how the ballot was cast**, across the 4 categories Georgia's counties tabulate in. And in 2020 the method is the story.

That is the sense in which this file is the right evidence for the questions below rather than merely an available one. Its known incompleteness is the recount, which the state published as a *separate* archive carrying 159 county summaries and precinct detail for exactly one county. The finest grain therefore exists only for the count that was later superseded, which is a fact about publication rather than about counting, and one this chapter returns to.

Cleaning here was mostly verification. The parser walks two different trees to the same shape. It reads each candidate's party out of the name string the county wrote, where (I) means incumbent rather than Independent and has to be stripped before the party is read. A one-character misreading there would have reassigned every incumbent in the file.

Nothing was imputed, meaning no missing figure was filled in by guessing, and no total was adjusted. What was done instead was to check the arithmetic the source claims. The 31,872 method rows add to 4,998,482 votes, and the precinct file's presidential total is 4,998,482, which is the same number. The precincts add to the county totals with nothing left over.

The 2024 JSON carries no “certified” flag anywhere in it, so it was tested the same way against Georgia’s certified statewide figures rather than trusted. The raw archives themselves are not committed here, because they are large and not ours to redistribute. If you want to rebuild this, the download instructions are printed, and the work stops rather than failing quietly when the archives are absent.

03 What a county’s tabulation export actually looks like

Nothing described so far arrives as a table. Here is Baker County, five precincts and the smallest county in the state, as it sits inside its own `detail.xml`. Attributes the parser does not use have been removed and the indentation narrowed to fit this page; every name and every number below is verbatim.

```
<ElectionResult>
  <Region>Baker</Region>
  <VoterTurnout totalVoters="2258" ballotsCast="1559">
    <Precincts>
      <Precinct name="Anna" totalVoters="166" ballotsCast="124"/>
      <Precinct name="Elmodel" totalVoters="401" ballotsCast="290"/>
      ...three more...
    </Precincts>
  </VoterTurnout>
  <Contest text="President of the United States" voteFor="1">
    <VoteType name="Undervotes" votes="4">
      <Precinct name="Anna" votes="0"/>
      ...
    </VoteType>
    <Choice text="Donald J. Trump (I) (Rep)" totalVotes="897">
      <VoteType name="Election Day Votes" votes="291">
        <Precinct name="Anna" votes="60"/>
        <Precinct name="Elmodel" votes="52"/>
        ...
      </VoteType>
      <VoteType name="Absentee by Mail Votes" votes="138">
        <Precinct name="Anna" votes="5"/>
        ...
      </VoteType>
    </Choice>
  </Contest>
</ElectionResult>
```

The votes are the leaves of a tree, not the cells of a table. A number sitting in that file knows nothing about itself. To learn that Anna cast 60 election-day votes for Trump, you have to remember four things from four different levels above it. The county from `Region`, the office from `Contest`, the candidate from `Choice`, and the method from `VoteType`. Flattening it means carrying all four down to the leaf and writing them beside the number. Each of the 159 counties is a separate file with a separate tree, and the median one holds 23 contests, of which this chapter wants one.

Four years later the same state published the same thing like this:

Field	Value
localResults[1].name	Baker County
...ballotItems[1].name	President of the US
...ballotItems[1].voteFor	1
...ballotOptions[1].name	Donald J. Trump (Rep)
...ballotOptions[1].politicalParty	Rep
...ballotOptions[1].voteCount	883
...groupResults[1].groupName	Election Day
...groupResults[1].voteCount	306
...groupResults[2].groupName	Absentee by Mail
...groupResults[2].voteCount	56
...precinctResults[1].name	Anna
...precinctResults[1].voteCount	165
...precinctResults[1].groupResults[1].groupName	Election Day
...precinctResults[1].groupResults[1].voteCount	73

Same state, same office, same county, same tree – and **not one of the names matches**.

The county is Baker in one and Baker County in the other. The office is President of the United States and President of the US. The method is Election Day Votes and Election Day. There is no version note anywhere saying this happened.

Here is what comes out. One row per precinct, county and precinct together as the key:

County	Precinct	Registered	Ballots cast	Trump	Biden	Presidential votes	Dem two-party %
Baker	Anna	166	124	93	31	124	25.00
Baker	Elmodel	401	290	141	148	289	51.21
Baker	Hoggards Mill	351	234	120	111	234	48.05
Baker	Milford	270	207	137	69	206	33.50
Baker	Newton	1,070	704	406	293	702	41.92

And, separately, one row per precinct per candidate per method – the column no other precinct file has, which exists only because the tree had a VoteType level in it:

County	Precinct	Candidate	Party	Method	Votes
Baker	Anna	Donald J. Trump	REP	Absentee by Mail Votes	5
Baker	Anna	Donald J. Trump	REP	Advanced Voting Votes	27
Baker	Anna	Donald J. Trump	REP	Election Day Votes	60
Baker	Anna	Donald J. Trump	REP	Provisional Votes	1
Baker	Anna	Joseph R. Biden	DEM	Absentee by Mail Votes	12

County	Precinct	Candidate	Party	Method	Votes
Baker	Anna	Joseph R. Biden	DEM	Advanced Voting Votes	7
Baker	Anna	Joseph R. Biden	DEM	Election Day Votes	12
Baker	Anna	Joseph R. Biden	DEM	Provisional Votes	0

Five precincts of tree became 5 rows in one file and 60 in the other. Statewide the two come to 2,656 and 31,872.

The decisions that got it there

One contest was kept out of every county's twenty-odd, and it was found by searching rather than by matching. The contest kept is the one whose name *contains* the word President, case-insensitively. An exact match would have been the obvious thing to write and it would have returned every precinct in 2020 and nothing at all in 2024, because President of the United States and President of the US are different strings. A loose match is doing real work here, and it is a bet: it would also have caught a *President of the Board of Education* had Georgia put one on the ballot.

Two spellings of the same four categories had to be declared identical. The 2020 archive names its methods with a suffix the 2024 export drops. Nothing in either file says they are the same categories; a reader has to look at them and decide, and the decision is what makes the two years comparable at all.

The same thing	In 2020	In 2024
the county	Baker	Baker County
the office	President of the United States	President of the US
how a ballot was cast	Election Day Votes	Election Day
the candidate	Donald J. Trump (I)(Rep)	Donald J. Trump (Rep)
registration per precinct	reported	absent

(I) means incumbent, not Independent. The party is read off the end of the candidate's own name, and one file writes an incumbent's name with that marker in it. Read left to right and the last thing in parentheses wins; read the first thing in parentheses and every sitting officeholder in Georgia changes party.

The key is the pair, never the precinct name. Baker is a county in south Georgia and also the first word of a precinct inside Cobb County. Precinct names are what the county typed, and they repeat across counties, so any join on the name alone silently merges places hundreds of miles apart.

Nothing was imputed and nothing was corrected. The 2024 file has no registration figures at all, and the blanks in the table above stay blank rather than being filled from anywhere else. Where the source could be checked, it was. The 31,872 method rows add to 4,998,482 votes, and the precinct file's presidential total is 4,998,482. That is the same number, arrived at down two different branches of the same tree.

That agreement is the only evidence there is that the flattening was done right. There is no independent precinct-level authority to check it against. There is only the file, checked against itself.

04 One row is one precinct in one county

County	Precinct	Registered	Ballots cast	Biden	Trump	Presidential votes	Dem two-party %
Appling	1B	1,309	995	116	870	993	11.76
Appling	1C	848	642	44	594	640	6.90
Appling	2	2,251	1,457	908	531	1,448	63.10

One row is one precinct in one county: registration, ballots cast, and votes for each presidential candidate. Nothing in it is derived and nothing is modeled; these are the numbers a county reported out of its own tabulation system.

05 What the whole file contains

What the source contains	Value
counties	159
precincts	2656
contest	President of the United States
candidates	3
vote methods	Absentee by Mail Votes Advanced Voting Votes Election Day Votes Provisional Votes
contests in the source (median county)	23
rows in vote_methods.csv	31872

2,656 precincts across 159 counties. And one column here does not exist in any other precinct dataset: **vote methods**. Georgia's counties tabulate and report separately by how the ballot was cast, so the file carries precinct × candidate × method. What that buys turns out to be considerable.

06 “Precinct” is not a standard size

Quantity	Value
Precincts with at least one presidential vote	2,653
Smallest	1
25th percentile	865
Median	1,629
75th percentile	2,528

Quantity	Value
Largest	15,531
Precincts recording zero presidential votes	3

The largest precinct in Georgia cast 15,531 presidential votes. The smallest cast 1. **And the extremes are not where the phrase “precinct size” suggests they are.**

County	Precinct	Presidential votes	Whole county is one precinct
Lumpkin	Dahlonega	15,531	yes
Coweta	Newnan Centre	15,134	no
Stephens	Senior Center	11,885	yes
Butts	Butts County Admin Bldg	11,771	yes
Jackson	Central Jackson	11,757	no
Jackson	West Jackson	11,693	no

The biggest precincts in Georgia are rural. The largest is Dahlonega, in Lumpkin County — one of the counties that reports the entire county as a single precinct. The largest precinct anywhere in the metro five (Fulton, DeKalb, Gwinnett, Cobb and Clayton) is 5,509, roughly a third of the state record.

County	Precincts	Presidential votes
Fulton	384	524,659
DeKalb	191	370,804
Gwinnett	156	413,865
Cobb	145	393,746
Bacon	1	4,668
Bleckley	1	5,706
Butts	1	11,771

16 counties report the whole county as a **single precinct**, carrying 87,616 votes between them with no internal geography at all. Meanwhile the five largest counties hold 36.4% of the state's precincts.

But metro precincts are not small either, which is where the intuitive reading fails twice over. Their median is 1,888 votes against 1,453 in the rest of the state. What a county like Fulton, with 384 precincts, does is *cap* the maximum — it is managing polling places, not drawing them to fit. So **the maximum is a fact about administration and the median is a fact about population**. Two different things are being called “precinct size”.

So **any per-precinct average across Georgia averages over units that are not comparable** — the “per what?” problem, one level finer than the jurisdiction problem in the election-administration data.

07 The state total, and one thing that does work

Candidate	Votes
Joseph R. Biden	2,474,507
Donald J. Trump	2,461,837
Jo Jorgensen	62,138
Total presidential votes	4,998,482
Two-party margin	12,670

Sum 2,656 precincts and Georgia's 2020 presidential result appears: **Joseph by 12,670 votes** out of 4,998,482 cast.

Precincts aggregate to counties exactly — every county total in `counties.csv` is reproduced by summing its precincts, with no residue. **Hold onto that.** Two later failures in this chapter are surprising only because this one works.

08 The column VEST never had

How the ballot was cast	Votes	% of all votes
Advanced Voting	2,694,879	53.9%
Absentee by Mail	1,316,943	26.3%
Election Day	975,540	19.5%
Provisional	11,120	0.2%

Only 19.5% of Georgia's 2020 presidential votes were cast in person on election day.

More than half arrived through "Advanced Voting", which is Georgia's in-person early voting, and roughly a quarter by mail.

A conventional precinct file gives you precinct × candidate. This gives you precinct × candidate × method, and in 2020 the method **is** the story.

09 Two electorates in the same state

How the ballot was cast	Joseph	Donald	Dem two-party %
Advanced Voting	1,250,509	1,419,161	46.8%
Absentee by Mail	849,729	451,157	65.3%
Election Day	367,205	587,697	38.5%
Provisional	7,064	3,822	64.9%

Biden won mail ballots 65.3% to 34.7%. Trump won election-day ballots 61.5% to 38.5%.

In a state decided by 12,670 votes out of five million, the two channels disagreed by roughly **27 percentage points**.

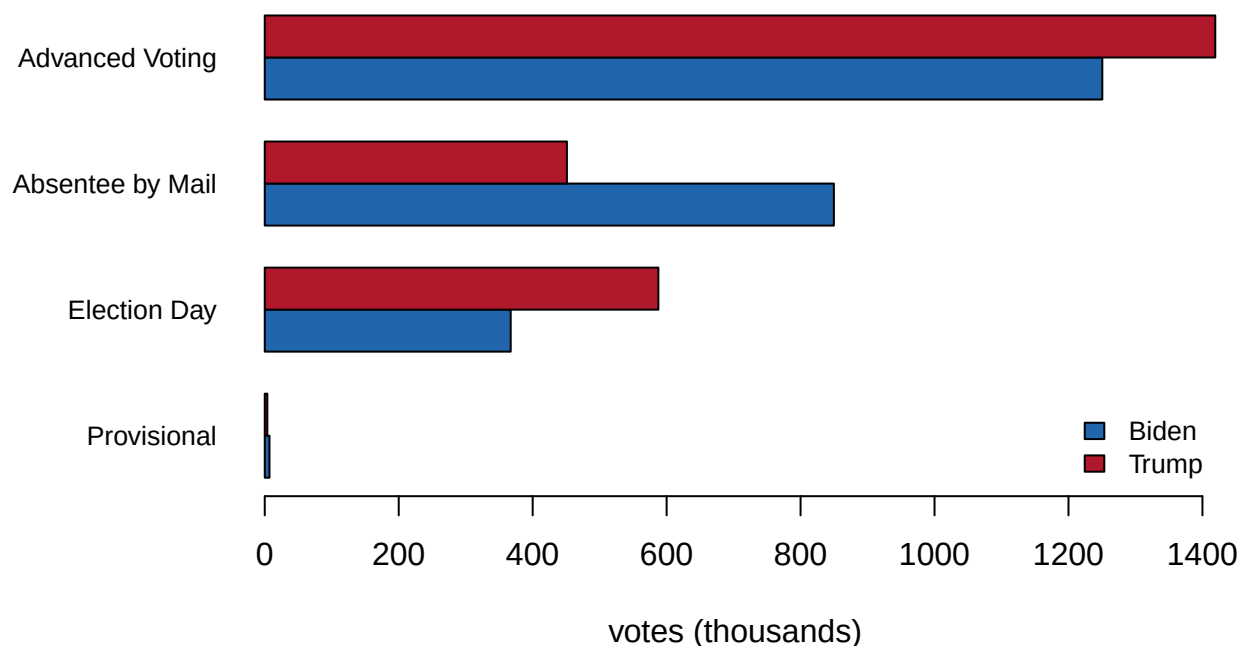


Figure 1. One electorate, four channels, and they do not agree. Georgia's 2020 presidential votes split by how the ballot was cast, with each candidate's total drawn as a bar and the Democratic two-party share printed at the end of each pair. Advanced Voting is the largest channel by a wide margin, carrying 53.9% of all votes. Election-day ballots, the route most people picture when they picture voting, carry 19.5%. Provisional ballots are the short pair at the bottom, 0.22% of the total. Across the four, the Democratic share runs from 38.5% to 65.3% — the same voters, sorted only by the route their ballot took.

If only these ballots counted	Two-party margin
Election Day only	Donald +220,492
Advanced Voting only	Donald +168,652
Absentee by Mail only	Joseph +398,572
All ballots	Joseph +12,670

This is the arithmetic behind the “red mirage”: counties that report election-day ballots first show one result, and the same counties reporting mail ballots later show another.

Nothing in that sequence is irregular. It is the same voters, sorted by how they chose to vote, and this file is where you can see it directly.

Before you read on, predict. You now have a complete, primary-source, precinct-level count of a presidential election decided by 12,670 votes. Georgia recounted it. Write down two numbers: **how many of the 159 counties you expect to have changed at all,** and **how much you expect the statewide margin to move.**

10 The same election, counted twice

Georgia published its November 2020 count and its November 2020 **recount** as two separate archives. Compare them.

Candidate	Original count	Recount	Change
Donald J. Trump	2,461,837	2,461,854	+17
Joseph R. Biden	2,474,507	2,473,633	-874

Quantity	Value
Counties in the comparison	159
Counties where the count changed	104
Share of counties	65.4%
Total votes moved (absolute)	2,753
Margin, original count	12,670
Margin, recount	11,779
Change in the margin	-891

104 of 159 counties returned a different number the second time — 65.4% of them. 2,753 votes moved in total, and the statewide margin narrowed by 891, from 12,670 to 11,779.

That second figure, **11,779**, is Georgia's certified 2020 margin. It is the number that was litigated, recounted again, and repeated in every account of the election. **The precinct-level file is the one that does not match it.**

This is the turn. The precinct file is the primary source, the finest grain, the complete 159-county publication — and it is *not* the official result. The official result comes from a different archive, produced by counting the same ballots a second time, which carries precinct detail for exactly **one** county. So the two can only be compared where the coarser file allows: at county level.

The lesson is not that somebody made a mistake. Recounts move numbers; that is what they are for, and a 0.055% movement across five million ballots is small. The lesson is that **"the count" is not a fact the data carries. It is a procedure the state performed, and it performed two of them.** Any precinct-level analysis you build rests on the count that was superseded, because it is the only count published at that resolution. That includes every racially polarized voting estimate in the second half of this book.

11 Where the movement was

County	Candidate	Original	Recount	Change	% change
Fulton	Joseph R. Biden	381,144	380,212	-932	-0.245
Floyd	Donald J. Trump	29,123	28,906	-217	-0.745

County	Candidate	Original	Recount	Change	% change
Gwinnett	Joseph R. Biden	241,827	241,994	+167	0.069
Henry	Joseph R. Biden	73,276	73,443	+167	0.228
Floyd	Joseph R. Biden	12,008	11,917	-91	-0.758
Muscogee	Joseph R. Biden	49,529	49,446	-83	-0.168

The movement is not spread evenly. **Fulton County alone accounts for 47.9% of all the change in Joseph's vote.** The largest county movements are in the largest counties, which is what you would expect if the recount was finding ordinary handling errors at a roughly constant rate per ballot, rather than anything concentrated.

12 The join that does not work

The aggregation earlier was perfect: precincts sum to counties, because counties drew the precincts inside their own borders.

Now try the join the rest of this course needs. Racially polarized voting requires returns and demographics **in the same place**. Returns come by precinct. Demographics come by census block.

Unit	Drawn by	Drawn when	Must respect the other?
Census block	The Census Bureau	Once a decade	No
Precinct	County election officials	Whenever the county needs to	No

They do not nest. A census block can straddle two precincts, and nothing requires either boundary to respect the other. So joining them means deciding how to split a block, and there is no correct answer, only conventions. Amos, McDonald and Watkins (2017) built the national version of exactly this crosswalk — a table that translates one set of areas into another. They documented what it costs in accuracy.

Every precinct-level demographic estimate in American politics rests on that allocation, including the ones that decide Voting Rights Act cases.

13 Precision and authority are different properties

What a precinct file gives that nothing coarser does is now on the table: 19.5% of Georgia's 2020 presidential votes were cast in person on election day. The mail and election-day electorates differed by about 27 points. And a 12,670-vote statewide margin sits on top of channels that disagreed that sharply. None of that is visible one level up.

The sharper claim is about counting. **Georgia counted the same ballots twice and returned different numbers in 65% of counties,** moving the statewide margin by 891 votes. And the precinct-level file is the superseded count, because the state never published the recount at precinct resolution.

What these files cannot do is also now clear. They do not say what any precinct's boundary looks like. They do not license comparing precincts across counties as though they were the same kind of unit. And they cannot be joined to census demographics without a splitting choice the data does not contain.

Which leaves the transferable finding, and it is not about Georgia. **Precision and authority are different properties.** This is the most precise election data available in the United States, and it disagrees with the official result. That is not because it is wrong. It is because "official" is a decision a state makes at a coarser resolution than the one it publishes.

14 What the argument is really about

The first argument is about who should be doing this work. For a decade the usable national precinct file was assembled by volunteers reading PDFs. That project is careful and it is not infallible, and there was no independent machine-readable source to check it against — which was the whole problem. The Georgia route used here is better and does not generalize: it exists because one state publishes structured county exports. Forty-nine other states do something else.

The second is about what a recount proves. Georgia's 2020 recount moved 2,753 votes and confirmed the outcome. To one reading that is reassurance: the count was accurate to 0.055% and the result held. To another it is evidence that hand counts and machine counts differ, which is the premise of a great deal of subsequent litigation. **Both readings use the same table.** The honest statement is the boring one — recounts move small numbers, and the margin here was much larger than the movement.

The third is about the block-to-precinct join. It is a methodological choice with no correct answer, it is invisible in every published result that depends on it, and it is load-bearing for expert testimony in redistricting cases.

15 What 2,656 precincts can testify to

The strength of this evidence is that it is the primary source rather than a reconstruction of one: the counties' own tabulation exports, published by the state, covering all 159 counties. It carries **vote method**, which almost no precinct dataset anywhere does. And it aggregates exactly to the published county totals, which is a real validation and not a formality.

Its limits each constrain a different sentence. **It is the superseded count.** The certified totals come from the recount archive, which carries precinct detail for exactly one county.

"Precinct" is not a comparable unit, and 16 counties report themselves as a single one. Precincts are identified by **names rather than stable identifiers**, so linking them across elections is manual work. There are **no boundaries** in these files; the shapes live elsewhere. There are **no demographics** either, and the join to them is a modeling choice rather than a lookup. And this is **one state and one election**: Georgia's vote-method reporting is unusual, and nothing here licenses assuming another state publishes it.

Three questions remain genuinely open. Which count is right is the first, and the least

answerable. The recount is certified and the precinct file is finer. Nobody reconciles them at precinct level, because the state never published one to reconcile.

How much the block-splitting convention moves later conclusions is testable, rarely tested, and consequential in litigation. And whether the method split reflects preference or access is beyond this file entirely. Mail and election-day ballots differed by 27 points, and nothing here separates who chose to vote by mail from who was unable to vote in person.

The chapter opened on a file with no stable keys in it, assembled from 159 county offices that were not coordinating. It ends on the same file disagreeing with the certified result of the election it describes, by 891 votes, across 104 counties. Neither of those is a scandal. Both of them are what it looks like when the most exact record available is not the authoritative one, which is the ordinary condition of American election data and almost never stated out loud.

16 Sources

Data

- Georgia Secretary of State, *November 3, 2020 General Election and November 3rd General Election Recount* — the counties' tabulation exports, all 159 counties. <https://sos.ga.gov/page/historical-elections-results> Download the election archive from that page; every table above is parsed out of it.
- Previously this chapter depended on the **Voting and Election Science Team's** precinct file via Harvard Dataverse (<https://doi.org/10.7910/DVN/NT66Z3>), which required a guestbook response. It no longer does.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`counties.csv`, `ga2020_vote_methods.csv`, `ga2024_structure.csv`, `precincts.csv`, `recount.csv`, `structure.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `recount.csv` has `original`, `recount` and `change` for every county and candidate. Sort by `pct_change`. Did the recount move votes in one direction, or scatter them?
- Add up the `change` column across all counties for each candidate. Did the recount change who won, and by how much did it change anything at all?
- `ga2020_vote_methods.csv` breaks every precinct's votes down by how they were cast. Pick a county and work out each candidate's share by method. How different are the election-day and mail electorates?
- `precincts.csv` has `registered` and `ballots_cast`. Compute turnout per precinct and sort. Find the extremes, then work out whether they are real or a boundary problem.

Academic literature

- Amos, B., McDonald, M. P. and Watkins, R. (2017). "When Boundaries Collide: Constructing a National Database of Demographic and Voting Statistics." *Public Opinion Quarterly* 81(S1): 385-400.

Further reading

- Ansolabehere, S. and Stewart, C. III (2005). "Residual Votes Attributable to Technology." *Journal of Politics* 67(2): 365-389.
- Hasen, R. L. (2020). *Election Meltdown: Dirty Tricks, Distrust, and the Threat to American Democracy*. Yale University Press.

WHY THIS DATA CANNOT BE FETCHED AGAIN

Part of the data behind this chapter comes from a source no assistant can fetch for you – though you can fetch it yourself, once, in a browser.

WHY NOT. Georgia's Secretary of State publishes one archive per election, covering all 159 counties down to the precinct, at <https://sos.ga.gov/page/historical-elections-results>

The address is stable and public and a browser downloads it fine, but the site sits behind bot protection: a script is turned away even with browser headers. The November 2020 general election and its recount are two separate archives there, and each is one manual browser download. Everything after that download is automatic. The finest-grained public record of American voting is served from behind a bot check, and that fact is part of the subject.

WHAT EXISTS INSTEAD. Download the two 2020 archives yourself from the page above. The 2024 general election needs no browser at all: the state's JSON export,

<https://results.sos.ga.gov/cdn/results/Georgia/export-2024NovGen.json> (about 34 MB), fetches cleanly by script. One more fact about what the state published: the recount archive carries county summaries for all 159 counties but precinct detail for exactly one county, Lanier – so the original count and the recount can only be compared county by county.

WHAT YOU CAN STILL CHECK. These numbers say whether you are holding the same objects:

- the 2020 archive yields 2,656 precincts across 159 counties

- the recount reproduces Georgia's certified 2020 totals exactly – Biden 2,473,633, Trump 2,461,854 – while the original precinct archive gives Biden 2,474,507 and Trump 2,461,837: the same election counted twice, 874 Biden votes apart
- the 2024 export sums to Trump 2,663,117 and Harris 2,548,017, matching the certified statewide figures